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Bayesian and likelihood inferences on remaining useful life in two-phase degradation models under gamma process

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ABSTRACT

Remaining useful life prediction has been one of the important research topics in reliability engineering. For modern products, due to physical and chemical changes that take place with usage and with age, a significant degradation rate change usually exists. Degradation models that do not incorporate a change point may not accurately predict the remaining useful life of products with two-phase degradation. For this reason, we consider the degradation analysis for products with two-phase degradation under gamma processes. Incorporating a probability distribution of the time at which the degradation rate changes into the degradation model, the remaining useful life prediction for a single product can be obtained, even though the rate change has not occurred during the inspection. A Bayesian approach and a likelihood approach via stochastic expectation-maximization algorithm are proposed for the statistical inference of the remaining useful life. A simulation study is carried out to evaluate the performance of the developed methodologies to the remaining useful life prediction. Our results show that the likelihood approach yields relatively less bias and more reliable interval estimates, while the Bayesian approach requires less computational time. Finally, a real dataset on LEDs is presented to demonstrate an application of the proposed methodologies.

1. Introduction

Due to advance technologies of sensors, there are multiple data sources in play. Therefore, prognostics and system health management (PHM) becomes an important topic in modern reliability engineering for improving the safety and performance of components and systems. In addition, there has been a growing interest of PHM in different fields including electronics, smart grid, nuclear plant, power industry, aerospace and military application, fleet industrial maintenance, and public health management [1]. PHM is a systematic approach for failure prevention by monitoring the health/status of products and systems, predicting failure progression, and mitigating operating risks through repair or replacement [1]. Effective prognostics that can yield an advance warning of impending failure in a system are critical for making maintenance decisions and executing preventive actions prior to failure occurrence to extend system life.

Systems are commonly installed with sensors that are designed to measure system performance. One of the important topics in PHM is degradation data analysis. Many papers have recently appeared on degradation models. Interested readers may refer to articles [2–5]. Wiener, gamma and inverse Gaussian processes are popular for modeling degradation data in the engineering literature. A Wiener process can represent increase and decrease in the performance of products over time, and thus is useful for some specific datasets. For monotonic degradation data, such as fatigue crack growth rates in metals of steel and aluminum alloy [6], resistance in carbon-film resistors [7], and light emitting diodes (LEDs) [8,9], gamma and inverse Gaussian processes are more appropriate for the degradation data analysis.

While the application of system health monitoring is established, degradation data that describe quality characteristics over time are collected. In degradation analysis, a failure time can be viewed as the first-passage time of a specified threshold in the degradation process. Furthermore, remaining useful life can be defined as the length of time from the current degradation to a pre-specified threshold, and has been frequently used as one of the vital indexes in PHM. For instance, remaining useful life prediction are valuable for condition-based maintenance policy. Si et al. [10] provided a comprehensive review on the statistical approaches of remaining useful life prediction. Chen and Tsui [11] presented a flowchart of model updating and remaining useful life prediction for condition-based maintenance. However, there is a lack of discussion on remaining useful life prediction under gamma process,

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probably due to the fact that the prediction has no closed form expression. In this article, an approximation using a two-parameter Birnbaum–Saunders distribution is presented.

In many existing research works on degradation models, it is common to assume that the degradation process is smooth and it does not have any change point. However, this strong assumption may be violated in practice. For instance, in studies of power outputs of lasers [12] and discharge capacity of a Li-ion battery [13], a significant change point is observed in the degradation. The degradation rate often increases due to physical and chemical changes that take place with usage and with age. Interested readers may refer to articles concerning degradation process with a change point such as [11,12,14,15]. However, those existing degradation models may not be appropriate for the analysis of light intensity of LEDs. Specifically, light intensity is usually monotonically decreasing in time. Without loss of physical reality, a gamma process with a change point is proposed in this paper for predicting the remaining useful life of products with two-phase degradation.

In a reliability study of lasers, the amounts of power outputs are measured at several inspection times, however, the time at which the degradation rate changes cannot be observed. To handle the data in presence of missing data, Ng [12] proposed an expectation-maximization (EM) algorithm [16] to find the maximum likelihood estimates. When the likelihood function under the proposed model can be complicated, a stochastic version on the EM algorithm, stochastic EM (SEM) algorithm [17], is an alternative for the parameter estimation. The SEM algorithm has recently been applied to many studies in reliability data analysis [18–20]. In this paper, the gamma process with a random change point is considered and a SEM algorithm is developed for the parameter estimation. On the other hand, the Bayesian approach is often adopted for parameter estimation. Many researchers applied Bayesian methods to numerous applications in degradation data analysis [4,11,15,21]. However, the choice of prior distributions is always one of the challenging tasks because inaccurate informative priors could seriously bias the parameter estimates, especially in the case of small sample sizes. One of the objectives in this paper is to compare the performance between the likelihood and Bayesian approaches in terms of bias and root mean square error of point estimation, and coverage probability of interval estimation of the remaining useful life.

The rest of this paper is organized as follows. In Section 2, a two-phase degradation model is formulated. For the purpose of PHM, the remaining useful life prediction is also presented in Section 2. In Section 3, a likelihood approach via SEM algorithm and a Bayesian approach are developed for estimation of model parameters. The maximum likelihood and the Bayesian inferences on the remaining useful life for a single product is presented in Section 4. In Section 5, a simulation study is carried out to evaluate the performance of the developed methodologies for different sample sizes and different numbers of inspections. In Section 6, the developed methodologies are illustrated by applying to analyze a real dataset on light emitting diodes (LEDs). Finally, in Section 7, some concluding remarks are made wherein some further issues of interest are pointed out.

2. A two-phase degradation model

A two-phase degradation model is motivated by data consisting of outputs of light intensity of twelve LEDs [22]. Fig. 1 displays the outputs of light intensity of the LEDs. It can be seen that the degradation rate changed around 500 hours. We can also observe that the outputs degraded rapidly in the initial phase and then degraded gradually. As the outputs are monotonically decreasing in time, it is therefore reasonable to formulate the outputs by using a two-phase degradation model under gamma process.

Suppose that the degradation measures of n items were obtained at times $0 = t_0 < t_1 < \ldots < t_m$. Let $Y_i(t_j)$ be the degradation measure of the ith item observed at time t_j, where $1 = Y_i(t_0) > Y_i(t_1) > \ldots > Y_i(t_m) \geq 0$ and let $\mathcal{T}$ be a random variable for the time at which the degradation rate changes, which is known as the change time. For instance, the ith item changes the degradation rate at τ_i, where τ_i is the actual value of the random variable $\mathcal{T}$. In this study, we assume that $\mathcal{T}$ is unobservable and it has a gamma distribution with shape parameter $u > 0$ and scale parameter $v > 0$ (denote as $Ga(u, v)$) with probability density function (PDF)

$$f_{\mathcal{T}}(\tau) = \frac{\tau^{u-1}}{\Gamma(u)v^u} \exp\left(-\frac{\tau}{v}\right), \quad \tau > 0. \tag{1}$$

Since the degradation measure is bounded between 0 and 1, a log-transformation is employed to the degradation measurements. We further denote $G_{i,j} = -\ln(Y_i(t_j))$ and $g_{i,j} = G_{i,j} - G_{i,j-1} > 0$. Given $\mathcal{T} = \tau_i$, it is assumed that $g_{i,j}$ has a gamma distribution with shape $\lambda_{i,j} > 0$ and scale $\beta > 0$, where

$$\lambda_{i,j} = \begin{cases} \alpha_1(t_j - t_{j-1}), & t_j \leq \tau_i, \\ \alpha_2(t_j - t_{j-1}), & t_{j-1} \geq \tau_i, \\ \alpha_1(\tau_i - t_{j-1}) + \alpha_2(t_j - \tau_i), & t_{j-1} < \tau_i < t_j. \end{cases} \tag{2}$$

The gamma process with an identical scale parameter β gives the important additive property. Suppose that the degradation between time t_1 and t_2 is $Ga(\lambda_1, \beta)$ and the degradation between time t_2 and t_3 is $Ga(\lambda_2, \beta)$. Then, based on the additive property, the degradation between time t_1 and t_3 is $Ga(\lambda_1 + \lambda_2, \beta)$. In general, the distribution of the degradation in between two time points always follows a gamma distribution. Therefore, the gamma process with an identical scale parameter is therefore widely used for maintenance planning [23]. This common scale parameter can also be viewed as the parameter that captures the time-independent degradation characteristics among all the item [24]. In addition, the linear shape function represents that the expected degradation is linear over time, and is also commonly used in reliability study [25,26].

In this case, the corresponding PDF of $g_{i,j}$ is

$$f_g(g|\mathcal{T} = \tau_i) = \frac{g^{\lambda_{i,j}-1}}{\Gamma(\lambda_{i,j})\beta^{\lambda_{i,j}}} \exp\left(-\frac{g}{\beta}\right), \quad g > 0. \tag{3}$$

Let X_i be the time-to-failure of the ith item, at which the degradation process crosses a pre-specified threshold δ, i.e.,

$$X_i = \inf\{x > 0 : Y_i(x) < \delta\}. \tag{4}$$

To predict the remaining useful life, given the current degradation, it is necessary to determine the conditional probability distribution of the length of time crossing the threshold from the current degradation. So, given that $Y_i(t_m) > \delta$, equivalently, $X_i > t_m$, conditional on $\mathcal{T} = \tau_i$, we have

$$D_i = -\ln(Y_i(x)) + \ln(Y_i(t_m)) \sim Ga(\lambda_i, \beta),$$

where

$$\lambda_i = \begin{cases} \alpha_1(x - t_m), & t_m < x \leq \tau_i, \\ \alpha_2(x - t_m), & \tau_i \leq t_m < x, \\ \alpha_1(\tau_i - t_m) + \alpha_2(x - \tau_i), & t_m < \tau_i < x. \end{cases}$$

D_i can be considered as a transformed increment between time t_m and $x > t_m$. Then, we readily obtain the conditional probability that the degradation crosses the threshold δ at time x, given the degradation $Y_i(t_m)$ at the current time t_m and the change time $\mathcal{T} = \tau_i$, as

$$\begin{aligned} F(x|t_m, Y_i(t_m), \delta, \mathcal{T} = \tau_i) &= \Pr(D_i > -\ln(\delta) + \ln(Y_i(t_m)), X_i < x|\mathcal{T} = \tau_i) \\ &= \int_{d_i}^{\infty} \frac{t^{\lambda_i - 1}}{\Gamma(\lambda_i)\beta^{\lambda_i}} \exp\left(-\frac{t}{\beta}\right) dt = 1 - \gamma\left(\lambda_i, \frac{d_i}{\beta}\right), \end{aligned}$$

where $d_i = -\ln(\delta) + \ln(Y_i(t_m))$ and $\gamma(s, w) = \int_0^w t^{s-1} \exp(-t) dt / \Gamma(s)$ is the lower incomplete gamma ratio.

Park and Padgett [25] approximated the distribution of X by using a two-parameter Birnbaum–Saunders distribution. In the proposed model

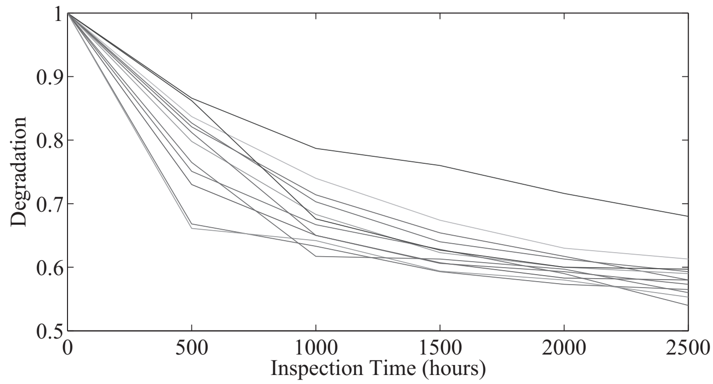

Fig. 1. Light intensity of twelve LEDs at electrical current of 40 mA.

here, given $\mathcal{T} = \tau_i$, it follows that λ_i has a Birnbaum–Saunders distribution with shape $\kappa_i = \sqrt{\beta/d_i}$ and scale $\eta_i = d_i/\beta$ (see [27–29]). Thus, we have

$$F(x|\mathcal{T} = \tau_i, t_m, Y_i(t_m)) \approx \Phi\left(\sqrt{\frac{d_i}{\beta}}\left(\sqrt{\frac{\lambda_i\beta}{d_i}} - \sqrt{\frac{d_i}{\lambda_i\beta}}\right)\right), \tag{5}$$

This result enables us to develop inferences on the remaining useful life, which is defined as the length from the current time t_m to the threshold δ based on the current degradation $Y_i(t_m)$. The expected remaining useful life for the ith item can be expressed as

$$\ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i) = E[X_i - t_m|Y_i(t_m), \delta, \mathcal{T} = \tau_i]. \tag{6}$$

In the remaining useful life prediction, the threshold is fixed and the failure time of a unit X_i is a random variable. If $X_i > \tau_i$, given $\mathcal{T} = \tau_i$, then λ_i is a function X_i, and thus has a Birnbaum–Saunders distribution with $E[\lambda_i|\mathcal{T} = \tau_i] = \eta_i(1 + \kappa_i^2/2) = d_i/\beta + 1/2$. Therefore,

$$\ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i) = \frac{-\ln(\delta) + \ln(Y_i(t_m))}{\alpha_2\beta} + \frac{1}{2\alpha_2} + \left(1 - \frac{\alpha_1}{\alpha_2}\right)\max\{0, \tau_i - t_m\}. \tag{7}$$

Since α_1 is the degradation rate in the initial phase and α_2 is the degradation rate in the second phase. It tells us that the performance of an item degraded rapidly, and subsequently degraded gradually for $\alpha_1 > \alpha_2$. From Eq. (7), it also agrees that, when the rate change occurred during the inspection, the remaining useful life prediction becomes larger for $\alpha_1 > \alpha_2$, while the remaining useful life prediction becomes smaller for $\alpha_1 < \alpha_2$.

If $X_i < \tau_i$, given $\mathcal{T} = \tau_i$, then λ_i has a truncated Birnbaum–Saunders distribution with shape parameter $\kappa_i = \sqrt{\beta/d_i}$ and scale parameter $\eta_i = d_i/\beta$ over the support $(0, \tau_i)$. Let Z be the standard normal distribution random variable, and $a_i = (\sqrt{\tau_i/\eta_i} - \sqrt{\eta_i/\tau_i})/\kappa_i$. It follows that

$$E[\lambda_i|X_i < \mathcal{T}, \mathcal{T} = \tau_i] = E\left[\eta_i\left(1 + \frac{\kappa_i^2 Z^2}{2} + \kappa_i Z\sqrt{1 + \frac{\kappa_i^2 Z^2}{4}}\right)\middle| Z < a_i, \mathcal{T} = \tau_i\right]$$
$$= \eta_i + \frac{\kappa_i^2\eta_i}{2}E_1 + \kappa_i\eta_i E_2, \tag{8}$$

where

$$E_1 = E\left[Z^2\middle|Z < a_i, \mathcal{T} = \tau_i\right]$$
$$= \left(\frac{1}{2} - \frac{a_i}{\sqrt{2\pi}}\exp\left(-\frac{a_i^2}{2}\right) + \frac{1}{2}erf\left(\frac{a_i}{\sqrt{2}}\right)\right)/\Phi(a_i), \tag{9}$$

$$E_2 = E\left[Z\sqrt{1 + \frac{\kappa_i^2 Z^2}{4}}\middle|Z < a_i, \mathcal{T} = \tau_i\right]$$
$$= \frac{1}{\Phi(a_i)}\left[-\frac{\sqrt{2\pi}\kappa_i}{4}\exp\left(\frac{2}{\kappa_i^2}\right)erfc\left(\frac{4 + \kappa_i^2 a_i^2}{\sqrt{2}\kappa_i}\right) - \frac{\sqrt{4 + \kappa_i^2 a_i^2}}{2}\exp\left(-\frac{a_i^2}{2}\right)\right], \tag{10}$$

$erf(x) = 2\int_0^x \exp(-t^2)dt/\sqrt{\pi}$ is the error function, and $erfc(x) = 1 - erf(x)$ is the complementary error function. The derivations of E_1 and E_2 are presented in the Appendix. Then, the remaining useful life prediction can be obtained as

$$\ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i) = \frac{1}{\alpha_1}\left(\eta_i + \frac{\kappa_i^2\eta_i}{2}E_1 + \kappa_i\eta_i E_2\right). \tag{11}$$

Moreover, given the rate change has not occurred during the inspection (i.e., $\tau_i > t_m$), the conditional probability that the degradation crosses the threshold with the change point is given by

$$p_\tau = \Pr(D_i < -\ln(\delta) + \ln(Y_i(t_m)), X_i < \tau_i|\mathcal{T} = \tau_i) = \gamma\left(\lambda_i, \frac{d_i}{\beta}\right), \tag{12}$$

where $d_i = -\ln(\delta) + \ln(Y_i(t_m))$ and $\lambda_i = \alpha_1(\tau_i - t_m)$.

3. Estimation of model parameters

In this section, we propose the likelihood and Bayesian approaches to estimate the model parameters in the two-phase degradation model described in Section 2

3.1. Likelihood approach via SEM algorithm

Let $\boldsymbol{\theta} = (\boldsymbol{\theta}_g, \boldsymbol{\theta}_\tau)$ be the model parameter vector, where $\boldsymbol{\theta}_g = (\alpha_1, \alpha_2, \beta)$ and $\boldsymbol{\theta}_\tau = (u, v)$. We further denote $\boldsymbol{\tau} = \{\tau_i, i = 1, 2, \ldots, n\}$ as the unobservable random rate change times, which can be viewed as the missing observations. The EM algorithm is an iterative procedure that repeatedly fills the missing data in the complete-data log-likelihood with their conditional expected values (E-step) and then maximizes the expected complete data log-likelihood to update the parameter estimates (M-step). In our case, the EM algorithm is hard to implement due to the random rate change times leading to complex forms of the conditional expectations in the E-step. Ye and Ng [18] stated that this difficulty in executing the E-step can be efficiently handled by the SEM algorithm by replacing the E-step with a stochastic step (S-step) as long as the missing data is easy to impute. We propose here the use of SEM algorithm for obtaining an approximation of the maximum likelihood estimates of the model parameters.

We first consider the log-likelihood function based on complete information given by

$$\ln L_c(\boldsymbol{\theta}) = \sum_{i=1}^{n}\sum_{j=1}^{m}\ln(f_g(g_{i,j}|\tau_i; \boldsymbol{\theta}_g)) + \sum_{i=1}^{n}\ln(f_{\mathcal{T}}(\tau_i; \boldsymbol{\theta}_\tau)). \tag{13}$$

In the rth iteration of the SEM algorithm, the objective is to maximize the function

$$Q(\theta|\theta^{(r)}) = E_{\theta^{(r)}}[\ln L_c(\theta)|\mathbf{g}] \tag{14}$$

with respect to θ at the M-step, and a value is randomly generated from the conditional distribution based on $\theta^{(r)}$ to replace each missing datum at the S-step. Then, a random sequence generated by SEM and the average of this random sequence after a burn-in period can serve as an approximation to the maximum likelihood estimate. To implement the SEM algorithm, given the previous parameter estimate $\theta^{(r)} = (\alpha_1^{(r)}, \alpha_2^{(r)}, \beta^{(r)}, u^{(r)}, v^{(r)})$, and the stochastically imputed missing data $\tau^{(r)}$, for $i = 1, 2, \dots, n$, the S-step can be described as follows:

A1. Compute

$$q\left(\theta^{(r)}, \tau_i^{(r)}\right) = \left[\prod_{j=1}^{m} f_g\left(g_{i,j}|\tau_i^{(r)}; \theta_g^{(r)}\right)\right] f_{\mathcal{T}}\left(\tau_i^{(r)}; \theta_\tau^{(r)}\right), \tag{15}$$

A2. Generate s_i from the uniform distribution in (0, 1) and set

$$\lambda_{i,k}^{(r)} = \begin{cases} \alpha_1^{(r)}(t_j - t_{j-1}), & j < k, \\ \alpha_2^{(r)}(t_j - t_{j-1}), & j > k, \\ \alpha_1^{(r)}(t_j - t_{j-1})s_i + \alpha_2^{(r)}(t_j - t_{j-1})(1 - s_i), & j = k. \end{cases} \tag{16}$$

A3. Compute

$$p_{i,k} = \prod_{j=1}^{m} f_g\left(g_{i,j}; \lambda_{i,k}^{(r)}, \beta^{(r)}\right)(F_{\mathcal{T}}(t_k) - F_{\mathcal{T}}(t_{k-1})), \tag{17}$$

for $k = 1, 2, \dots, m+1$, where we define $t_{m+1} \equiv \infty$.

A4. Generate K_i from a multinomial distribution with probabilities $p_{i,k}/\sum_{k=1}^{m+1} p_{i,k}$, $k = 1, 2, \dots, m+1$.

A5. Generate τ_i^* from truncated gamma distribution with $u^{(r)}$ and $v^{(r)}$ over the interval $[t_{K_i-1}, t_{K_i}]$.

A6. Compute $p_i = \min\left(1, q\left(\theta^{(r)}, \tau_i^*\right)/q\left(\theta^{(r)}, \tau_i^{(r)}\right)\right)$,

A7. Set $\tau_i^{(r+1)} = \begin{cases} \tau_i^* & \text{with probability} \quad p_i, \\ \tau_i^{(r)} & \text{with probability} \quad 1 - p_i. \end{cases}$

In the M-step of the SEM algorithm, we obtain the updated parameter estimate $\theta^{(r+1)} = \left(\theta_g^{(r+1)}, \theta_\tau^{(r+1)}\right)$ by maximizing the log-likelihood function

$$\begin{aligned} \ln L_c(\theta) &= \sum_{i=1}^{n}\sum_{j=1}^{m} \ln\left[f_g\left(g_{i,j}|\tau_i^{(r+1)}; \theta_g\right)\right] + \sum_{i=1}^{n} \ln\left[f_{\mathcal{T}}\left(\tau_i^{(r+1)}; \theta_\tau\right)\right] \\ &= \ln L_1(\theta_g) + \ln L_2(\theta_\tau) \end{aligned} \tag{18}$$

with respect to θ. Taking the first derivatives of the log-likelihood function in Eq. (18) with respect to u and v and setting them to zero yields

$$v = \frac{\sum_{i=1}^{n} \tau_i}{nu} \tag{19}$$

and

$$\psi(u) - \ln(u) = \frac{\sum_{i=1}^{n} \ln(\tau_i)}{n} - \ln\left(\frac{\sum_{i=1}^{n} \tau_j}{n}\right). \tag{20}$$

Since there is no explicit solution for u in Eq. (20), the function *fsolve* in Matlab is used to solve Eq. (20) to obtain the estimate of u. Then, the estimate of v can be obtained by substituting the estimate of u into Eq. (19).

Similarly, taking the first derivative of the log-likelihood function in Eq. (18) with respect to β and setting it to zero yields

$$\beta = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} g_{i,j}}{\sum_{i=1}^{n}\sum_{j=1}^{m} \lambda_{i,j}} = \frac{-\sum_{i=1}^{n} \ln(Y_i(t_m))}{\sum_{i=1}^{n}\sum_{j=1}^{m} \lambda_{i,j}}. \tag{21}$$

Substituting Eq. (21) into $\ln L_1(\theta_g)$, we can obtain

$$\sum_{i=1}^{n}\sum_{j=1}^{m} (\lambda_{i,j} - 1)\ln(g_{i,j}) - \ln(\Gamma(\lambda_{i,j})) - \lambda_{i,j} \ln\left(\frac{-\sum_{i=1}^{n} \ln(Y_i(t_m))}{\sum_{i=1}^{n}\sum_{j=1}^{m} \lambda_{i,j}}\right)$$

$$+ \frac{g_{i,j} \sum_{i=1}^{n}\sum_{j=1}^{m} \lambda_{i,j}}{\sum_{i=1}^{n} \ln(Y_i(t_m))}. \tag{22}$$

Once again, the quantity is a nonlinear function of α_1 and α_2, the function *fminsearch* in Matlab is used to obtain the solution of Eq. (22), which yields the estimates of α_1 and α_2.

We repeat the S- and M-steps N times to obtain sequences of $\theta^{(r)}$ and $\tau^{(r)}$, for $r = 1, 2, \dots, N$, and discard the first $\lceil N/10 \rceil$ iterations as burn-in, where $\lceil a \rceil$ denotes the ceiling function of a. We further denote the sequences after burn-in as $\theta^{[b]}$ and $\tau^{[b]}$, for $b = 1, 2, \dots, N - \lceil N/10 \rceil$. Then, the estimates of the parameter vector θ and the rate change time τ based on the proposed SEM algorithm are, respectively,

$$\hat{\theta} = \frac{1}{N - \lceil N/10 \rceil} \sum_{b=1}^{N-\lceil N/10 \rceil} \theta^{[b]}$$

and

$$\hat{\tau} = \frac{1}{N - \lceil N/10 \rceil} \sum_{b=1}^{N-\lceil N/10 \rceil} \tau^{[b]}.$$

3.2. Bayesian approach

In the Bayesian framework, the unknown model parameters α_1, α_2, β, u and v are assumed to be random, independent and have their own prior distributions. For statistical inference purpose, we are interested in the posterior distribution

$$\pi(\theta|\mathbf{g}, \tau) \propto \frac{\pi(\mathbf{g}|\theta, \tau)\pi(\theta_g)\pi(\tau|\theta_\tau)\pi(\theta_\tau)}{\pi(\tau|\mathbf{g})}, \tag{23}$$

where $\pi(\theta_g)$ and $\pi(\theta_\tau)$ are the prior distributions of $\theta_g = (\alpha_1, \alpha_2, \beta)$ and $\theta_\tau = (u, v)$, respectively,

$$\pi(\mathbf{g}|\theta, \tau) = \prod_{i=1}^{n}\prod_{j=1}^{m} f_g(g_{i,j}|\tau_i; \theta_g), \tag{24}$$

$$\pi(\tau|\theta_\tau) = \prod_{i=1}^{n} f_{\mathcal{T}}(\tau_i; \theta_\tau), \tag{25}$$

and

$$\pi(\tau|\mathbf{g}) = \prod_{i=1}^{n} f_{\mathcal{T}|g}(\tau_i|g_{i,j}, j = 1, 2, \dots, m), \tag{26}$$

with

$$f_{\mathcal{T}|g}(\tau_i|g_{i,j}, j = 1, 2, \dots, m) = \frac{\prod_{j=1}^{m} f_g(g_{i,j}|\tau_i; \theta_g) f_{\mathcal{T}}(\tau_i; \theta_\tau)}{\int_0^\infty \prod_{j=1}^{m} f_g(g_{i,j}|\tau_i; \theta_g) f_{\mathcal{T}}(\tau_i; \theta_\tau) d\tau_i}.$$

The following Markov Chain Monte Carlo (MCMC) sampling scheme via Metropolis–Hastings algorithm [30] is used to sample from the posterior distribution $\pi(\theta|\mathbf{g}, \tau)$ in the rth step:

B1. Generate $\theta_g^* = (\alpha_1^*, \alpha_2^*, \beta^*)$ from a distribution based on $\theta_g^{(r)}$ and $\theta_\tau^* = (u^*, v^*)$ from a distribution based on $\theta_\tau^{(r)}$.

B2. Compute $p = \min\left(1, \prod_{i=1}^{n} q\left(\theta^*, \tau_i^{(r+1)}\right)\pi(\theta^*)/\prod_{i=1}^{n} q\left(\theta^{(r)}, \tau_i^{(r+1)}\right)\pi(\theta^{(r)})\right)$, where $\theta^* = (\theta_g^*, \theta_\tau^*)$

B3. Set $\theta^{(r+1)} = \begin{cases} \theta^* & \text{with probability } p \\ \theta^{(r)} & \text{with probability } 1 - p \end{cases}$

Suppose we obtain $M(>>1000)$ samples from the posterior distribution as $\theta^{(r)}$, for $r = 1, 2, \dots, M$, then we discard the first 1000 samples as burn-in and sample one value in every 100 iterations after the burn-in. Hence, we have $B = \lfloor (M - 1000)/100 \rfloor$ samples from the posterior distribution, where $\lfloor a \rfloor$ denotes the floor function of a. Let $\theta^{[b]}$, $b = 1, 2, \dots, B$, be the sample from the posterior distribution, the Bayesian estimate of the parameter vector θ can be obtained as

$$\tilde{\theta} = \frac{1}{B}\sum_{b=1}^{B}\theta^{[b]}.$$

Since the conditional distribution $f_{\mathcal{T}|g}(\tau_i|g_{i,j}, j = 1, 2, \dots, m)$ provides the unit-specific distribution of the rate change time of unit i, we can use the sample average of the sample τ_i in the above sampling process as an estimate of the rate change time of unit i, i.e.,

$$\tilde{\tau}_i = \frac{1}{B}\sum_{b=1}^{B}\theta^{[b]},$$

where $\tau_i^{[b]}, b = 1, 2, \dots, B$ is the sample from the conditional distribution $f_{\mathcal{T}|g}(\tau_i|g_{i,j}, j = 1, 2, \dots, m)$ obtained by discarding the first 1000 samples as burn-in and sample one value in every 100 iterations after the burn-in.

4. Inference on remaining useful life

With the sequences of the model parameter vector and the rate change times based on either the likelihood approach via SEM algorithm or the Bayesian approach, we can generate sequences of the degradation and the corresponding expected remaining useful life using the following steps:

C1. Set $d_i = -\ln(\delta) - G_{i,m}$. When $\tau_i^{[b]} < t_m$, compute the expected remaining useful life $\mu_{\ell_i}^{[b]} = \ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i^{[b]})$ by using (7), and generate a remaining useful life $\ell_i^{[b]}$ from Birnbaum–Saunders distribution with shape $\kappa_i = \sqrt{\beta/d_i}$ and scale $\eta_i = d_i/\beta/\alpha_2$.

C2. When $\tau_i^{[b]} > t_m$, compute p_τ by using (12), and generate a Bernoulli random variable with p_τ to determine whether the rate change occurred before the failure time.

C2-a. If the rate change occurred before the failure time, compute the expected remaining useful life $\mu_{\ell_i}^{[b]} = \ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i^{[b]})$ by using (7), generate $d_{i2} = -\ln(Y_i(t_m)) + \ln(Y_i(\tau_i^{[b]})$ from a truncated gamma distribution with scale $\alpha_1(\tau_i^{[b]} - t_m)$ and scale β over (0, d_i), and generate a remaining useful life $\ell_i^{[b]}$ from Birnbaum–Saunders distribution with shape $\kappa_i = \sqrt{\beta/(d_i - d_{i2})}$ and scale $\eta_i = (d_i - d_{i2})/\beta/\alpha_2$.

C2-b. If the rate change occurred after the failure time, compute the expected remaining useful life $\mu_{\ell_i}^{[b]} = \ell(t_m, Y_i(t_m), \delta|\mathcal{T} = \tau_i^{[b]})$ by using (11), and generate a remaining useful life $\ell_i^{[b]}$ from truncated Birnbaum–Saunders distribution with shape $\kappa_i = \sqrt{\beta/d_i}$ and scale $\eta_i = d_i/\beta/\alpha_1$ over $(0, \tau_i^{[b]})$.

When the sequences of $\boldsymbol{\mu}_{\ell}^{[b]}$ and $\boldsymbol{\ell}^{[b]}$ are obtained, the point estimate of the remaining useful life for the ith item is given by

$$\hat{\mu}_{\ell_i} = \frac{1}{B}\sum_{b=1}^{B}\mu_{\ell_i}^{[b]}. \tag{27}$$

A $100(1-\alpha)\%$ Bayesian credible interval for the remaining useful life of the ith item, ℓ_i, can be expressed as

$$\left(\ell_i^{[\lfloor B\alpha/2\rfloor]}, \ell_i^{[\lceil B(1-\alpha/2)\rceil]}\right). \tag{28}$$

Based on the parameter estimates obtained from SEM algorithm, a percentile bootstrap technique can be used for constructing a confidence interval for the remaining useful life:

D1. Simulate a bootstrap degradation data, $\boldsymbol{g}_{Boot}$, based on the parameter estimates $\hat{\theta}$.

D2. Obtain the bootstrap estimate θ^* based on $\boldsymbol{g}_{Boot}$ via SEM algorithm and hence the point estimate of the remaining useful life, say $\hat{\mu}_{\ell_i}^{(l)}$.

D3. Repeat C1 and C2 L times to obtain $\hat{\mu}_{\ell_i}^{(l)}$ and $\ell_i^{(l)}$, $l = 1, 2, \dots, L$.

D4. Order $\hat{\mu}_{\ell_i}^{(l)}$ and $\ell_i^{(l)}$, $l = 1, 2, \dots, L$ in ascending order to obtain $\hat{\mu}_{\ell_i}^{[1]} < \hat{\mu}_{\ell_i}^{[2]} < \dots < \hat{\mu}_{\ell_i}^{[L]}$ and $\ell_i^{[1]} < \ell_i^{[2]} < \dots < \ell_i^{[L]}$.

The point estimate of the remaining useful life for the ith item is given by

$$\hat{\mu}_{\ell_i} = \frac{1}{L}\sum_{l=1}^{L}\mu_{\ell_i}^{[l]}. \tag{29}$$

A $100(1-\alpha)\%$ bootstrap confidence interval for the remaining useful life for the ith item, ℓ_i, can be expressed as

$$\left(\ell_i^{[\lfloor L\alpha/2\rfloor]}, \ell_i^{[\lceil L(1-\alpha/2)\rceil]}\right). \tag{30}$$

It is noting that the SEM algorithm and the Bayesian approach for parameter estimation are similar in the sense that they both generate a sequence of estimates. However, in the Bayesian approach, the sequence of estimates is also used for the interval estimation of the remaining useful life, while the sequence of estimates obtained by SEM is used for the approximation of the maximum likelihood estimates and a bootstrap technique is then implemented for the interval estimation of the remaining useful life.

5. Simulation study

In this section, an extensive Monte Carlo simulation study is conducted to examine the performance of the proposed point and interval estimation methods in terms of their biases, root mean square errors (RMSE), coverage probabilities (CP) and computational time, for different sample sizes $n = (12, 30)$ and different numbers of inspection $m = (10, 20)$. Suppose that the degradation of each item was measured at $(m+1)$ different times, $t_j = 5j, j = 0, 1, \dots, m$. A failure is defined when the percentage of degradation to the initial degradation is below 70%, thus δ is set to be 0.7. We consider $\boldsymbol{\theta}_g = (5/3, 1/3, 0.001), \boldsymbol{\theta}_\tau = (8, 4)$, $N = 1000$ and $M = 10000$.

The data generating scheme for the degradation measure $Y_i(t_j)$ at time t_j and the failure time X_i of ith item is as follows:

1. Generate τ_i from a gamma distribution with scale u and shape v, for $i = 1, 2, \dots, n$.
2. Compute $\lambda_{i,j}$ from (2), $i = 1, 2, \dots, n, j = 1, 2, \dots, 300$.
3. Generate $g_{i,j}$ from a gamma distribution with scale β and shape $\lambda_{i,j}$, for $i = 1, 2, \dots, n, j = 1, 2, \dots, 300$.
4. Compute $Y_i(t_j) = \exp\left(-\sum_{j=1}^{m} g_{i,j}\right)$.
5. Set $X_i = \underset{t_j=\{5,10,\dots,1500\}}{\arg\min}\ Y_i(t_j) < \delta$.

Since the family of gamma distributions is a conjugate family with respect to the gamma likelihood function with unknown scale parameter [23] and the unknown parameters in the gamma distributions are positive [4], any statistical distribution with non-negative support can be used as the prior distribution of $\boldsymbol{\theta}_g$ and $\boldsymbol{\theta}_\tau$. In our study, we assume that the prior distribution $\pi(\boldsymbol{\theta}_g)$ are gamma distributions with shape parameters (500/3, 100/3, 0.1) and scale parameters (0.01, 0.01, 0.01) and the prior distribution $\pi(\boldsymbol{\theta}_\tau)$ are gamma distributions with shape parameters (4, 2) and scale parameters (2, 2), respectively. As suggested by Singpurwalla and Wilson [24], the hyperparameters may be assigned by noting the means and standard deviations of $\boldsymbol{\theta}_g$ and $\boldsymbol{\theta}_\tau$. Thus, in the Step B1 presented in Section 3.2, $\boldsymbol{\theta}_g^*$ and $\boldsymbol{\theta}_\tau^*$ are generated from gamma distributions with expected values of $\theta_g^{(r)}$ and $\theta_\tau^{(r)}$, respectively and scale parameter 0.001. For the bootstrap confidence interval based on the estimate obtained from SEM algorithm, we use $L = 1000$ bootstrap samples. The results obtained from the simulation study, based on 1000 Monte Carlo simulations, are summarized in Tables 1– 2. These results show that the estimate of the remaining useful life becomes more accurate when the number of inspection increases. As expected, increasing the sample size n or increasing the number of inspection m reduces the

Table 1
Biases, RMSEs and coverage probabilities (CP) of the remaining useful life and the change time for the sample sizes $n = 12$ and 30, and for the number of inspections $m = 10$ and 20.

	remaining useful life				change time			
	$m = 10$		$m = 20$		$m = 10$		$m = 20$	
$n = 12$	Bayesian	SEM	Bayesian	SEM	Bayesian	SEM	Bayesian	SEM
Bias	36.688	6.652	20.431	3.217	−0.301	0.179	−0.273	0.214
RMSE	69.162	85.510	48.873	48.226	1.040	1.080	0.706	0.752
CP	0.944	0.973	0.933	0.955	0.863	0.940	0.876	0.951
$n = 30$	Bayesian	SEM	Bayesian	SEM	Bayesian	SEM	Bayesian	SEM
Bias	36.444	0.396	19.503	−0.819	−0.242	−0.158	−0.062	0.025
RMSE	59.593	52.240	36.348	31.030	0.686	0.675	0.439	0.416
CP	0.911	0.967	0.931	0.950	0.863	0.937	0.871	0.945

Table 2
Comparison of computation times (in seconds) between Bayesian and SEM for the sample sizes $n = 12$ and 30, and for the number of inspections $m = 10$ and 20.

	$m = 10$		$m = 20$	
	Bayesian	SEM	Bayesian	SEM
$n = 12$	145.3	1823.2	164.6	2023.8
$n = 30$	259.8	2979.8	290.3	3443.4

RMSEs of the point estimates. In addition, the SEM algorithm yields relatively more reliable interval estimates while the Bayesian approach requires less computational time.

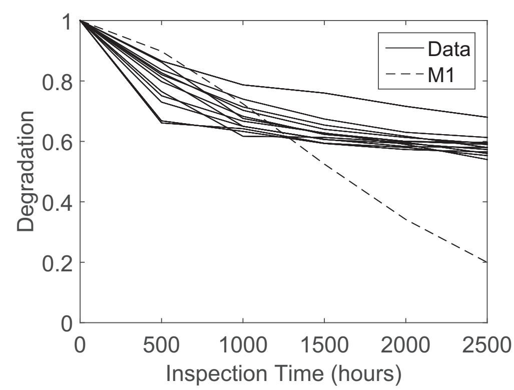

Fig. 2. Gamma process without degradation rate change (M1).

6. Illustrative example

As an illustration, we perform the degradation analysis on the light intensity data of twelve LEDs presented in Fig. 1 using the proposed two-phase degradation model. For the Bayesian approach, we assume that the prior distributions $\pi(\theta_g)$ and $\pi(\theta_\tau)$ are gamma distributed with shape parameters (2.54, 0.51, 1.95) and (56, 1.6), and scale parameters (0.01, 0.01, 0.01) and (10, 1), respectively. In Step B1 of the Bayesian computation, $\boldsymbol{\theta}_g^*$ and $\boldsymbol{\theta}_\tau^*$ are sampled from gamma distributions with expected values of $\theta_g^{(r)}$ and $\theta_\tau^{(r)}$, respectively, and scale parameter 0.001. A failure is defined as the light intensity degrades below a threshold of 50% of the initial value.

Table 4 presents the remaining useful life prediction for each LED under gamma process without (M1) and with (M2) degradation rate change, along with the corresponding 95% bootstrap confidence intervals (with $L = 1000$) or 95% Bayesian credible interval, based on SEM algorithm with $N = 10000$ and Bayesian approach with $M = 100000$. Fig. 3 displays the degradation of three LEDs, along with the corresponding 95% bootstrap confidence interval and 95% Bayesian credible interval. From Fig. 3, we observe that the degradation rate in the initial phase is significantly larger than that in the second phase. Moreover, Table 3 reveals that both likelihood and Bayesian approaches yield similar estimates for the model parameters $\boldsymbol{\theta}_g = (\alpha_1, \alpha_2, \beta)$ but different estimates for $\boldsymbol{\theta}_\tau = (u, v)$. Hence, from Fig. 4, the PDF and the estimates of the remaining useful life of the three LEDs are slightly different. Table 4 and Fig. 2 reveal that the remaining useful life prediction under gamma process without degradation rate change is considerably underestimated.

If one is interested in the degradation rate change time of each LED, we can obtain the estimate of the expected value and the corresponding quantiles based on the methods described in Section 3. The results are presented in Table 5.

7. Concluding remarks

In this paper, we proposed a two-phase degradation model that incorporates the random rate change time in a gamma process. Both likelihood and Bayesian approaches are developed for parameter estimation and the remaining useful life prediction for an individual item under the proposed two-phase degradation model. The proposed model and methodologies are applied to analyze the data of outputs of light intensity of 12 LEDs and subsequently make statistical inference on the remaining useful life of each LED.

Table 3
Estimates of the model parameters under gamma process without (M1) and with (M2) degradation rate change, along with the corresponding 95% credible intervals or 95% bootstrap confidence intervals.

Model	M1	M2			
	Likelihood	Bayesian		SEM	
Parameter	Estimate	Estimate	Credible interval	Estimate	Bootstrap CI
α_1	0.0027	0.020	(0.016, 0.025)	0.020	(0.013, 0.030)
α_2	-	0.004	(0.003, 0.005)	0.004	(0.003, 0.006)
β	0.0787	0.024	(0.020, 0.031)	0.027	(0.018, 0.038)
u	-	548	(535, 565)	566	(235, 1308)
v	-	1.182	(0.985, 1.394)	1.600	(0.576, 3.158)

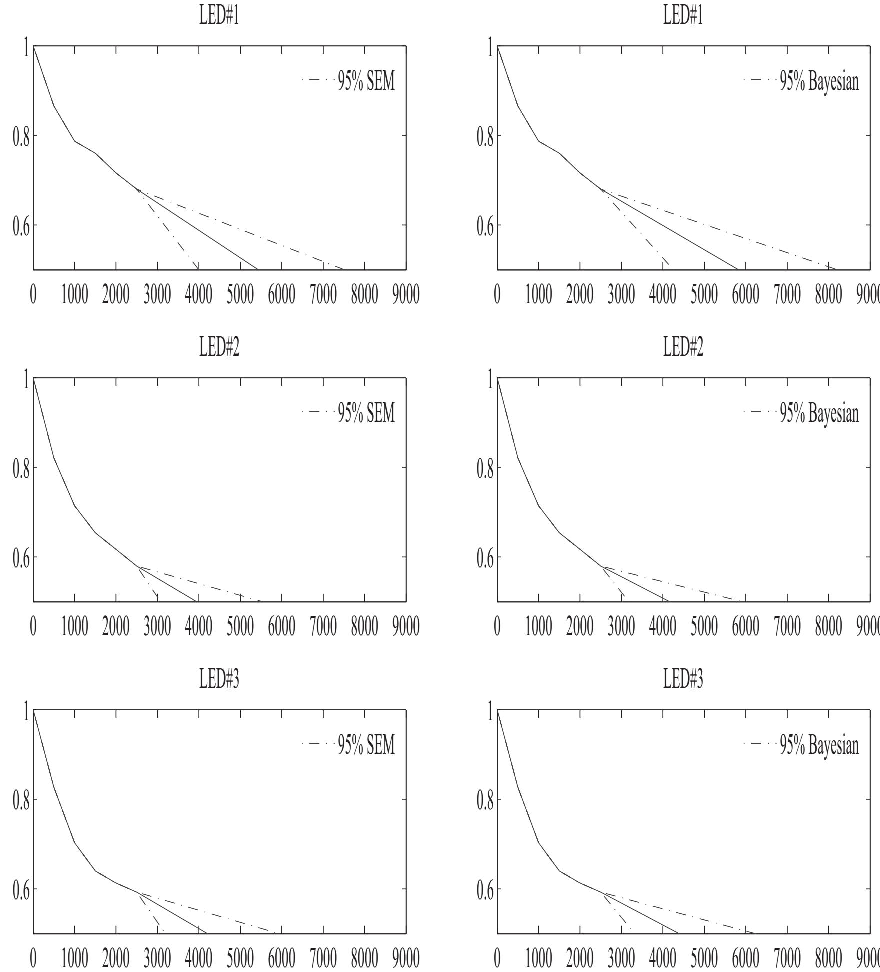

Fig. 3. Remaining useful life of LEDs (#1–3), along with the corresponding 95% confidence intervals, through the stochastic EM algorithm (left) and the Bayesian approach (right).

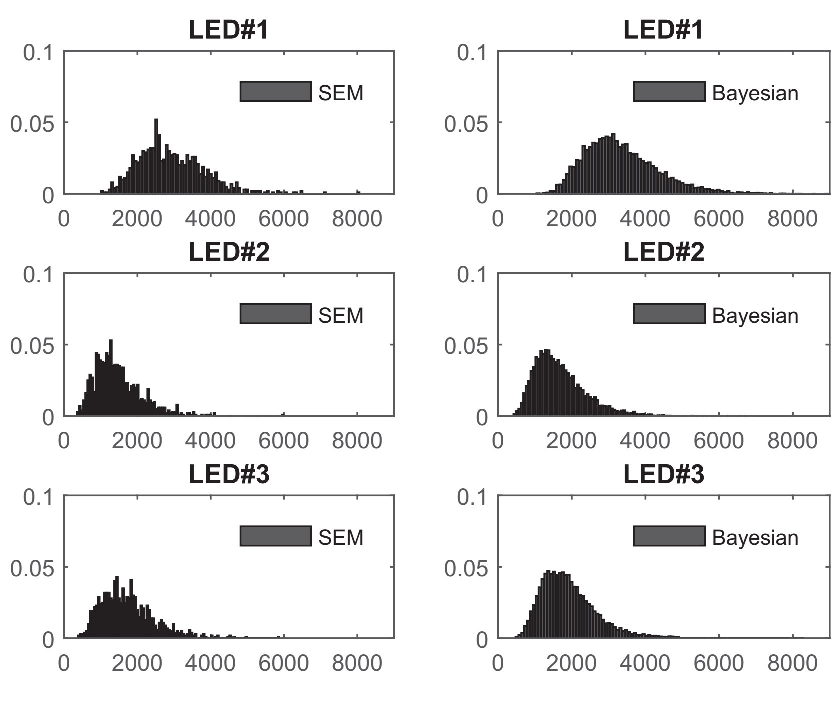

Fig. 4. The histograms of estimates of remaining useful life of LEDs (#1–3), through the Bayesian approach (left) and the stochastic EM algorithm (right).

Table 4
Remaining useful life prediction for each LED under gamma process without (M1) and with (M2) degradation rate change, along with the corresponding 95% credible intervals or 95% bootstrap confidence intervals.

Model		M1	M2			
		Likelihood	Bayesian		SEM	
Unit ID	$Y(2500)$	Estimate	Estimate	Credible interval	Estimate	Bootstrap CI
1	0.680	0.0120	3247.4	(1726.0, 6213.9)	2925.1	(1499.9, 4942.5)
2	0.580	0.0065	1658.8	(671.9, 3268.3)	1467.1	(571.7, 3059.9)
3	0.593	0.0073	1873.3	(900.7, 3035.3)	1731.6	(694.0, 3461.6)
4	0.590	0.0071	1781.5	(729.5, 3330.5)	1644.5	(693.1, 3339.6)
5	0.540	0.0040	910.0	(301.6, 2008.3)	819.9	(224.0, 2127.0)
6	0.613	0.0084	2190.0	(1156.4, 4197.8)	2026.6	(921.7, 4089.1)
7	0.580	0.0065	1638.8	(507.1, 3048.3)	1525.4	(556.3, 3289.6)
8	0.597	0.0075	1870.8	(626.8, 4141.9)	1734.1	(742.3, 3586.8)
9	0.573	0.0061	1641.1	(539.2, 3694.0)	1399.5	(527.5, 2927.7)
10	0.565	0.0056	1338.0	(518.3, 3068.4)	1224.7	(442.7, 2776.3)
11	0.553	0.0049	1219.4	(394.8, 2739.3)	1051.4	(337.7, 2666.9)
12	0.560	0.0053	1222.6	(447.3, 2682.1)	1149.4	(410.9, 2633.5)

Table 5
Estimate the change time of each LED, along with the corresponding 2.5% and 97.5% quantiles.

Unit ID	Bayesian		SEM	
	Estimate	Quantiles	Estimate	Quantiles
1	648	(536, 777)	753	(686, 826)
2	644	(552, 778)	753	(681, 822)
3	650	(542, 791)	752	(682, 826)
4	650	(549, 767)	751	(683, 829)
5	651	(543, 777)	754	(688, 832)
6	645	(546, 764)	751	(686, 818)
7	652	(563, 798)	754	(683, 826)
8	655	(556, 786)	754	(684, 827)
9	653	(537, 768)	752	(682, 828)
10	646	(539, 765)	754	(682, 824)
11	639	(533, 756)	749	(685, 824)
12	652	(550, 784)	754	(686, 822)

A Monte Carlo simulation study is conducted and shows that both the likelihood and Bayesian approaches successfully yield accurate prediction on the remaining useful life. In addition, the proposed estimation approaches perform satisfactory even when the rate change times are unobservable in the data collection process. Moreover, it reveals that the PDF of the estimates of the remaining useful life is well approximated through the SEM algorithm and leads to reliable interval estimates. However, the SEM algorithm involves intensive computation to find the estimates of the model parameters, and therefore is recommended for off-line degradation analysis to obtain an accurate prediction.

Gamma distributions can be used to describe an increasing or decreasing hazard rate. In addition, its additive property suggests that $\mathcal{T}$ has a gamma distribution in this study, which leads to an approximation of the remaining useful life prediction by using Birnbaum–Saunders distribution. It is also noting that the scale parameter β is assumed to be the same over time in this study, due to the additive property of gamma distribution that the sum of independent gamma random variables having the same scale parameter is also a gamma random variable. This assumption eases the computation of the remaining useful life of an individual item. Indeed, this assumption can be released to a more general situation that different items have different scale parameters over time. However, the number of model parameters required to be estimated will extensively increase, which leads to the identifiability issue when the number of items under monitoring is small. When the number of items is under monitoring is large and all the model parameters can be estimated, the computational time and effort will increase significantly when different items have different scale parameters over time in the gamma process. We are currently working on the generalization of the proposed model and hope to report the findings in a future paper.

In M-step, the likelihood function involves the change point $\mathcal{T}$ that is unobservable from the degradation test. We can obtain the information about the change point from the conditional distribution of $\mathcal{T}$ in the estimation procedure. The proposed procedure also allows us to determine whether the degradation rate change has occurred during the test. If the degradation rate change has not occurred during the test, the conditional distribution of $\mathcal{T}$ can provide some insights on the change to the remaining useful life prediction.

Table I shows that the choice of the prior distribution has a considerable influence on the posterior distribution and hence the results of the Bayesian inference, especially when the sample size is small. The choice of prior distribution is always a critical issue in using the Bayesian method. Therefore, the prior sensitivity analysis [30] should be conducted to evaluate the impact of prior distributions on the posterior inference. Peng et al. [4] also discussed a reliable posterior analysis within a Bayesian framework. It is of a great interest to investigate the goodness-of-fit tests and to compare different prior distributions for degradation model with a change point. Research in these directions are in progress and we hope to report the results in a future paper.

Our numerical results were obtained in Matlab on a desktop with an Intel Core i3-4130 processor, a 3.40 GHz CPU and 4GB of memory. Matlab code for simulation and the data used in Section 4 is available upon request from the authors.

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Appendix

$$E_1 = E\left[Z^2 \middle| Z < a_i, \mathrm{T} = \tau_i\right] \tag{31}$$

$$= \int_{-\infty}^{a_i} \frac{x^2}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) dx / \Phi(a_i) \tag{32}$$

$$= \left(\frac{1}{2} - \frac{a_i}{\sqrt{2\pi}} \exp\left(-\frac{a_i^2}{2}\right) + \frac{1}{2} erf\left(\frac{a_i}{\sqrt{2}}\right) \right) / \Phi(a_i) \tag{33}$$

$$E_2 = E\left[Z\sqrt{1 + \frac{\kappa_i^2 Z^2}{4}} \middle| Z < a_i, \mathrm{T} = \tau_i \right] \tag{34}$$

$$= \int_{-\infty}^{a_i} \frac{x}{\sqrt{2\pi}} \sqrt{1 + \frac{\kappa_i^2 x^2}{4}} \exp\left(-\frac{x^2}{2}\right) dx / \Phi(a_i) \tag{35}$$

$$= \frac{1}{\sqrt{2\pi}} \left(\int_0^{a_i^2/2} \sqrt{1 + \frac{\kappa_i^2 y}{2}} \exp(-y) dy - \int_0^{\infty} \sqrt{1 + \frac{\kappa_i^2 y}{2}} \exp(-y) dy \right) / \Phi(a_i) \tag{36}$$

$$= \left(-\frac{\sqrt{2\pi}\kappa_i}{4} \exp\left(\frac{2}{\kappa_i^2}\right) erfc\left(\frac{4 + \kappa_i^2 a_i^2}{\sqrt{2}\kappa_i}\right) - \frac{\sqrt{4 + \kappa_i^2 a_i^2}}{2} \exp\left(-\frac{a_i^2}{2}\right) \right) / \Phi(a_i) \tag{37}$$

The posterior distribution is

$$\pi(\boldsymbol{\theta}|\boldsymbol{g}, \boldsymbol{\tau}) = \frac{\pi(\boldsymbol{\theta}, \boldsymbol{g}, \boldsymbol{\tau})}{\pi(\boldsymbol{g}, \boldsymbol{\tau})} = \frac{\pi(\boldsymbol{g}|\boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\theta}, \boldsymbol{\tau})}{\pi(\boldsymbol{\tau}|\boldsymbol{g})\pi(\boldsymbol{g})}$$

$$\propto \frac{\pi(\boldsymbol{g}|\boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\tau}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\pi(\boldsymbol{\tau}|\boldsymbol{g})} = \frac{\pi(\boldsymbol{g}|\boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\tau}|\boldsymbol{\theta}_\tau)\pi(\boldsymbol{\theta}_g)\pi(\boldsymbol{\theta}_\tau)}{\pi(\boldsymbol{\tau}|\boldsymbol{g})}.$$

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2017.11.017.

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